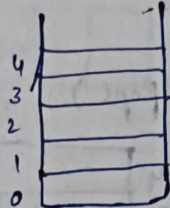


1. perform the following operations in LIFO data structure having an array size of 5. Push (45), Push (22), Push (11), Push (77), Pop(), Push (44), Push (33), Pop(), Pop(), Pop() Print the peak element of the data structure and its index position with a neat sketch explained in detail

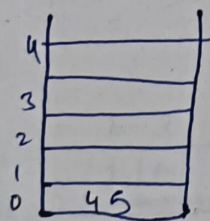
Solution

an array size of 5



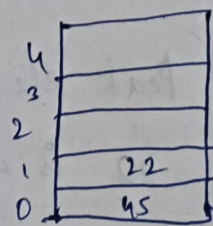
27
36

(i) push (45)



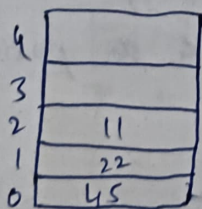
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= -1 + 1 \\ &= 0 \end{aligned}$$

(ii) push (22)



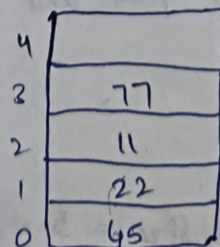
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

(iii) push (11)



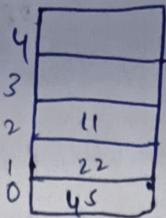
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 1 + 1 \\ &= 2 \end{aligned}$$

iv) Push (77)



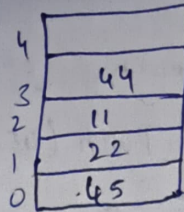
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

v) pop()



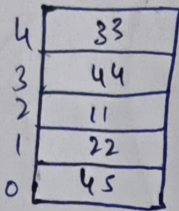
$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

vi) push(44)



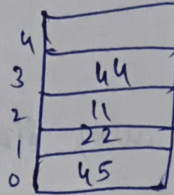
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 2 + 1 \\ \text{top} &= 3 \end{aligned}$$

vii) push(33)



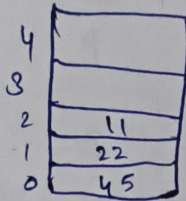
$$\begin{aligned} \text{top} &= \text{top} + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

viii) pop()



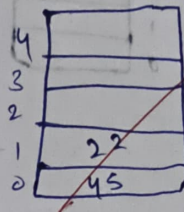
$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 4 - 1 \\ &= 3 \end{aligned}$$

ix) pop()



$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 3 - 1 \\ &= 2 \end{aligned}$$

x) pop()



$$\begin{aligned} \text{top} &= \text{top} - 1 \\ &= 2 - 1 \\ \text{top} &= 1 \end{aligned}$$

Peak element = 22

22 is at position 1.

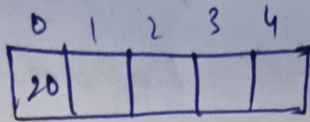
2. Perform the following operations in Circular queue having a size of 2.
- Enqueue(20), Enqueue(10), Enqueue(30), Dequeue(), Enqueue(80), Dequeue(), Enqueue(90), Enqueue(100), Dequeue(), Dequeue(), Enqueue(60), Enqueue(90), Finally display the front and rear elements with its positions. Discuss in detail.

Solution

$$\text{Size} = 5$$

$$\text{Front} = \text{rear} = -1$$

Enqueue(20)

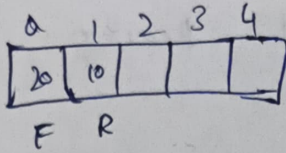


FR

$$F=0, R=0$$

$$\text{Que}(\text{Rear}) = 20$$

ii) Enqueue(10)



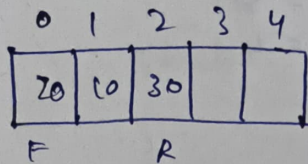
$$F=0, R=1$$

$$\text{Que}(\text{Rear}) = 10$$

$$\text{Rear} = \text{Rear} + 1 = 0 + 1$$

$$\text{Rear} = 1$$

iii) Enqueue(30)



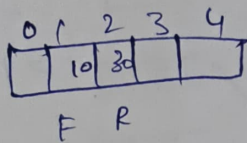
$$\text{Que}(\text{Rear}) = 30$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 1 + 1$$

$$\text{Rear} = 2 \quad F = 0$$

iv) Dequeue()

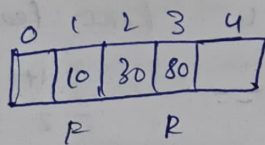


$$\text{Front} = \text{Front} + 1$$

$$= 0 + 1$$

$$\text{Front} = 1 \quad R = 2$$

v) Enqueue(80)

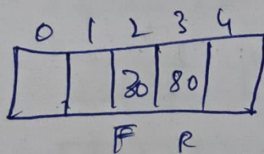


$$\text{Front} = 1 \quad \text{Rear} = \text{Rear} + 1$$

$$= 2 + 1$$

$$= 3$$

vi) Dequeue()



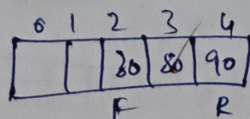
$$\text{Front} = \text{Front} + 1$$

$$\text{Rear} = 3$$

$$= 1 + 1$$

$$= 2$$

vii) Enqueue(90)

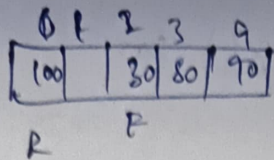


$$\text{Front} = 2 \quad \text{Rear} = \text{Rear} + 1$$

$$= 3 + 1$$

$$= 4$$

vii) Enqueue(100)

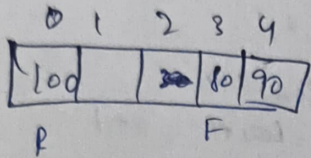


Queue is full

$$\text{Front} = 2$$

$$\text{Rear} = 0 - 1 + 1 = 0$$

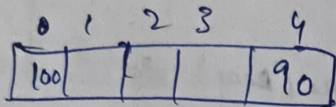
ix) Dequeue()



$$\begin{aligned} \text{Front} &= \text{Front} + 1 \\ &= 2 + 1 \\ &= 3 \end{aligned}$$

$$\text{Rear} = 0$$

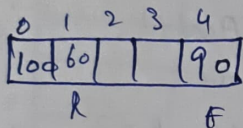
x) Dequeue()



$$\begin{aligned} \text{Front} &= \text{Front} + 1 \\ &= 3 + 1 \\ &= 4 \end{aligned}$$

$$\text{Rear} = 0$$

xii) Enqueue(60)



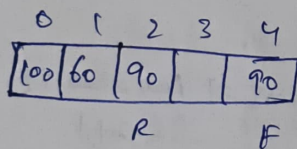
$$\text{Front} = 4$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 0 + 1$$

$$= 1$$

xiii) Enqueue(90)



$$\text{Front} = 4$$

$$\text{Rear} = \text{Rear} + 1$$

$$= 1 + 1$$

$$= 2$$

Front element 90 at position '4'

Rear element 90 at position '2'

Convert the following infix expression into a postfix expression.

Solution.

$$B - (C/D + (E \cdot (F * G)/H)^*) J$$

Also evaluate the given postfix expressions. a) 9 3 8 4 / * +

b) 6 2 + 6 3 / *

$$B - (C/D + (E \cdot (F * G)/H)^*) J$$

Solution

S.C stack Postfix expression

B

--	--

 B

-

-	
---	--

 B

C

-	C	
---	---	--

 B

C

-	C	C
---	---	---

 BC

D

-	C	C	D
---	---	---	---

 BC

/

-	C	C	/
---	---	---	---

 BCD

+

-	C	+
---	---	---

BCD/

C

-	C	+	C
---	---	---	---

 BCD/ ~~BCD~~

E

-	C	+	C	E
---	---	---	---	---

 BCD/E

·/

-	C	+	C	·/
---	---	---	---	----

 BCD/E

F

-	C	+	C	·/	F
---	---	---	---	----	---

 BCD/EF

*

-	C	+	C	·/	*
---	---	---	---	----	---

BCD/EF

G

-	C	+	C	·/	*	G
---	---	---	---	----	---	---

 BCD/EF G

J

-	C	+	C	·/	*	G	J
---	---	---	---	----	---	---	---

)

-	C	+
---	---	---

 BCD / EF . / . G *

/

-	C	+
---	---	---

 BCD / EF . / . G *

H

-	C	+	1
---	---	---	---

 BCD / EF . / . G * H

)

-	
---	--

 BCD / EF . / . G * H /

*

*	
---	--

 BCD / EF . / . G * H / + -

J.

*	
---	--

 BCD / EF . / . G * H / + - J

OVER

--

Empty

BCD / EF . / . G * H / + - J → output

a) 9384 / * +

The

9

9	
---	--

3

9	3
---	---

8

9	3	8
---	---	---

4

9	3	8	4
---	---	---	---

/

9	3	2
---	---	---

 → $8 \div 4 = 2$

*

9	6
---	---

 → $9 + 6 = 15$

+

15

O/P = 15

b) 62 + 63 / *

6

6

2

6	2
---	---

+

8

8

8	6
---	---

3

8	6	3
---	---	---

*

8	2
---	---

+

16

$6 + 2 = 8$

$6 \div 3 = 2$

O/P = 16